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Sample Question Set - Unit 4

Course Code: CSE252
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
Course Outcome: CO4
Paper: Sample Multiple Choice Questions - ANSWER KEY
Maximum Marks: 40 x 1 = 40

Unit: 4
Unit Title: Unsupervised Learning
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the branch of machine learning that works on data carrying no target column.

A) Unsupervised learning <-- correct
B) Supervised learning
C) Reinforcement learning
D) Transfer learning

Answer: A. With no correct answer recorded for any row, the task becomes finding the structure that is already present in the features.

Q2. Identify what the value of k specifies in the k-Means algorithm.

A) The number of features to be used
B) The number of clusters to be produced <-- correct
C) The number of iterations to be run
D) The number of records in each cluster

Answer: B. k must be supplied before the algorithm runs, which is why methods such as the elbow and the silhouette exist to help choose it.

Q3. State the name of the diagram produced by hierarchical clustering.

A) A scree plot
B) A confusion matrix
C) A dendrogram <-- correct
D) A scatter matrix

Answer: C. The dendrogram records the order of the merges and the height at which each one occurred, and a horizontal cut across it gives a partition.

Q4. Name the DBSCAN parameter that fixes how many neighbours a point needs in order to be a core point.

A) eps
B) k
C) The linkage criterion
D) MinPts <-- correct

Answer: D. eps sets the radius of the neighbourhood and MinPts sets how many points must lie inside it. Both are needed to define a core point.

Q5. Identify the quantity that the first principal component captures more of than any other direction.

A) Accuracy
B) Variance <-- correct
C) Correlation with the target
D) The number of records

Answer: B. The components are the directions of greatest spread, produced in decreasing order of the variance each accounts for.

Q6. State what the confidence of an association rule measures.

A) The share of all transactions holding both items
B) The number of items appearing in the rule
C) The number of passes the algorithm required

D) The share of antecedent transactions that also hold the consequent <-- correct

Answer: D. Support is measured over all transactions; confidence is measured only over those that contain the antecedent.

Q7. Explain why the result of a clustering cannot be reported as an accuracy figure.

A) There is no known correct grouping to compare it against <-- correct

B) The clusters are always of unequal sizes

C) The distance measure is chosen after the run

D) The dataset is always too small to score

Answer: A. Accuracy needs a known answer for each record. Internal measures such as the silhouette, and profiles of the clusters, are reported instead.

Q8. Distinguish k-Means from hierarchical clustering by whether the number of clusters must be supplied.

A) Hierarchical needs it beforehand, k-Means does not

B) Both of the two need it supplied beforehand

C) k-Means needs it beforehand, hierarchical does not <-- correct

D) Neither of the two needs it supplied at all

Answer: C. Hierarchical clustering builds the entire tree of merges, and the number of clusters is decided afterwards by where the cut is placed.

Q9. Describe what a robot uses unsupervised learning on its sensor data for.

A) Predicting a labelled fault code for each reading

B) Choosing the actions that earn the largest reward

C) Grouping readings into operating modes and spotting unusual ones <-- correct

D) Converting the readings into written language

Answer: C. No labels accompany raw sensor streams, so grouping similar conditions together and flagging what does not fit are the natural tasks.

Q10. Interpret a Principal Component Analysis in which the first two components explain 45 and 30 per cent of the variance.

A) The two together account for 75 per cent of the spread <-- correct

B) The two together account for 13.5 per cent

C) The first component is correlated 0.45 with the second

D) The dataset must have had exactly two features

Answer: A. Cumulative explained variance is the running total, and it is the usual basis for deciding how many components to retain.

Q11. Predict the effect of one extreme outlier on a k-Means centroid.

A) The centroid is unaffected by the outlier

B) The outlier is automatically discarded

C) The cluster is split into two new clusters

D) The centroid is pulled towards the outlier <-- correct

Answer: D. A centroid is a mean, and a mean has no resistance to an extreme value. Removing or capping such records before clustering is common practice.

Q12. Recognize the task described: readings from a machine are grouped into operating modes with no labels supplied.

A) Classification

B) Clustering <-- correct

C) Regression

D) Reinforcement learning

Answer: B. The groups are discovered from the readings themselves, which is exactly the unsupervised setting.

Q13. Explain why the features must be standardised before Principal Component Analysis when their units differ.

A) Standardisation removes the outliers from the data

B) Standardisation makes the components correlated

C) Standardisation reduces the number of components

D) Variance depends on the unit, so one column would dominate <-- correct

Answer: D. A pressure reading in pascals has a far larger numeric spread than a temperature in degrees, and the components would describe the pressure alone.

Q14. Summarize what association rule mining produces from transaction data.

A) A numeric prediction for each transaction

B) A cluster label for each transaction

C) Rules stating that items tend to appear together <-- correct

D) A reduced set of features for each item

Answer: C. The rules are then filtered by support, confidence and lift, and acted upon through shelf placement, bundling or cross-selling.

Q15. Distinguish the support of a rule from its confidence.

A) Confidence is over all transactions, support over the antecedent's

B) Support is over all transactions, confidence over the antecedent's <-- correct

C) Both of the two are measured over all transactions

D) Both are measured over the antecedent's transactions

Answer: B. A rule can have low support because the combination is rare, yet high confidence because it holds almost whenever the antecedent appears.

Q16. Explain why DBSCAN can label a point as noise while k-Means cannot.

A) DBSCAN grows clusters from dense regions and leaves the rest <-- correct

B) DBSCAN is given the number of clusters in advance

C) k-Means measures density rather than distance

D) k-Means requires the features to be standardised

Answer: A. k-Means assigns every record to its nearest centre, so a stray point must join some cluster. DBSCAN sets aside anything that is not in a dense region.

Q17. Predict what happens to the inertia of a k-Means result as k is increased.

A) It rises steadily without limit

B) It falls steadily towards zero <-- correct

C) It remains constant whatever k is

D) It rises and then falls again

Answer: B. Each extra centre lets points sit closer to a centre, reaching zero when every record is its own cluster, which is why inertia alone cannot choose k.

Q18. Describe what the elbow method plots.

A) Inertia against the number of clusters <-- correct

B) Accuracy against the training size

C) Support against the confidence

D) Variance against the component number

Answer: A. The value of k at which the curve visibly bends is taken as the choice, on the grounds that further clusters buy little further reduction.

Q19. Explain why the labels a clustering assigns to its groups are arbitrary.

A) They are ordered by the size of each group

B) They record the distance to the cluster centre

C) They match the true classes of the records

D) They are names for the groups, not meaningful values <-- correct

Answer: D. The same partition with the labels renamed is the same result, which is why an external measure compares which records are grouped together, not the names.

Q20. Distinguish dimensionality reduction from feature selection.

A) Selection builds new combined features, reduction keeps originals

B) Both of the two build new combined features

C) Reduction builds new combined features, selection keeps original ones <-- correct

D) Both of the two keep only the original features

Answer: C. This is why selection is preferred when each retained feature has to be explained in the terms it was measured in.

Q21. Apply k-Means with k set to 2 to the one-dimensional points 2, 4, 20 and 22, and determine the two centroids at convergence.

A) 3 and 21 <-- correct

B) 2 and 22

C) 12 and 12

D) 4 and 20

Answer: A. The natural grouping is {2, 4} and {20, 22}, whose means are 3 and 21 respectively.

Q22. Solve the following: a database holds 600 transactions and an itemset appears in 90 of them. Determine its support.

A) 0.09

B) 0.60

C) 0.15 <-- correct

D) 1.5

Answer: C. Support is the share of all transactions containing the itemset: 90 / 600 = 0.15.

Q23. Use the following counts to determine the confidence of the rule that if A occurs then B occurs: A appears in 200 transactions and A with B in 150.

A) 0.15

B) 0.75 <-- correct

C) 1.33

D) 0.25

Answer: B. Confidence is the share of antecedent transactions holding the consequent: 150 / 200 = 0.75.

Q24. Apply Principal Component Analysis to a dataset whose components explain 60, 20, 12 and 8 per cent, and determine how many are needed for at least 90 per cent.

A) 2

B) 4

C) 1

D) 3 <-- correct

Answer: D. The cumulative shares are 60, 80 and 92 per cent, so three components pass the threshold.

Q25. Compute the percentage reduction in width when a dataset of 40 features is reduced to 10 components.

A) 25 per cent

B) 10 per cent

C) 75 per cent <-- correct

D) 40 per cent

Answer: C. The width falls by 30 of the original 40 columns, so 30 / 40 = 0.75, a reduction of 75 per cent.

Q26. Examine the following and determine the correct explanation: k-Means run twice on the same data with the same k reports two different partitions.

A) The dataset changed between the two runs

B) The distance measure changed between runs

C) The value of k was silently altered

D) The runs started from different random centres <-- correct

Answer: D. The objective is not convex, so each run settles in the local minimum its starting centres lead to. Multiple restarts and a recorded seed are the usual response.

Q27. Compare k-Means with DBSCAN on the shape of cluster each is able to recover.

A) k-Means recovers convex groups, DBSCAN any dense shape <-- correct

B) DBSCAN recovers convex groups, k-Means any dense shape

C) Both of the two recover only convex groups

D) Both of the two recover clusters of any shape

Answer: A. Each k-Means cluster is the set of points nearest one centre, which is always convex. DBSCAN follows connected dense regions however they curve.

Q28. Apply the Apriori property to an itemset holding A and B with a support of 0.2, and determine the largest possible support of the itemset holding A, B and C.

A) 0.4

B) 0.2 <-- correct

C) 0.8

D) 1.0

Answer: B. Every transaction containing the larger set must contain the smaller one, so adding an item can never raise the support above 0.2.

Q29. Use a dendrogram cut that crosses three vertical lines and determine the number of clusters obtained.

A) 1

B) 2

C) 6

D) 3 <-- correct

Answer: D. Each vertical line the cut crosses is one cluster surviving at that height.

Q30. Solve the following: a cluster holds the one-dimensional points 6, 10 and 14. Determine its centroid.

A) 6

B) 10 <-- correct

C) 14

D) 30

Answer: B. The centroid is the mean of the members: (6 + 10 + 14) / 3 = 30 / 3 = 10.

Q31. Apply DBSCAN with MinPts set to 4 to a point that has 3 neighbours within eps and lies near no core point, and determine its status.

A) A core point

B) A border point

C) A noise point <-- correct

D) A cluster centre

Answer: C. It fails the density test itself and is not reachable from a point that passes it, so it belongs to no cluster.

Q32. Compute the lift of a rule whose support is 0.2, with an antecedent support of 0.4 and a consequent support of 0.4.

A) 1.25 <-- correct

B) 0.5

C) 2.0

D) 0.8

Answer: A. Lift is 0.2 / (0.4 x 0.4) = 0.2 / 0.16 = 1.25, so the pair occurs slightly more often than chance would predict.

Q33. Examine a clustering of raw sensor data holding temperature in degrees and pressure in pascals, and determine what governs the result.

A) The pressure column, because of its far larger range <-- correct

B) The temperature column, because it is measured first

C) Both columns contribute equally to the distance

D) Neither column, since the clustering fails to run

Answer: A. Squared differences in pascals swamp those in degrees, so the partition is effectively made on pressure alone until the columns are standardised.

Q34. Use Principal Component Analysis on a dataset of three features and determine the maximum number of components it can produce.

A) 2

B) 1

C) 6

D) 3 <-- correct

Answer: D. The components form a new basis for the same space, so there can be at most as many of them as there were original features.

Q35. Apply hierarchical clustering to 15 records and determine the number of merges before a single cluster remains.

A) 15

B) 14 <-- correct

C) 16

D) 7

Answer: B. Each merge reduces the count of clusters by one, so 15 singletons require 15 - 1 = 14 merges.

Q36. Compare the inertia of a k-Means result at k equal to 1 with the inertia at k equal to 5 on the same data.

A) It is lower at k equal to 1

B) It is identical at both values

C) It is lower at k equal to 5 <-- correct

D) It cannot be computed at k equal to 1

Answer: C. Inertia falls with every extra cluster, which is why the shape of the curve rather than its value is what the elbow method reads.

Q37. Examine a rule reported with a confidence of 0.9 and a lift of 1.0, and determine what should be concluded.

A) The rule is among the strongest in the dataset
B) The confidence must have been miscalculated
C) The consequent is common and the rule adds nothing <-- correct
D) The antecedent almost never occurs at all
Answer: C. A lift of one means the two items are independent. The high confidence merely reflects how often the consequent appears on its own.

Q38. Use the standard procedure to determine the number of clusters for a robot grouping unlabelled terrain readings.

A) Set k to the number of features in the data
B) Try several values of k and compare elbow and silhouette <-- correct
C) Set k to the number of records in the data
D) Set k to two, since that is always sufficient
Answer: B. No label exists to check against, so the number has to be argued from internal evidence and from whether the groups mean anything in the application.

Q39. Apply a minimum support of 0.05 to a database of 2,000 transactions and determine the corresponding count.

A) 50
B) 200
C) 400
D) 100 <-- correct
Answer: D. The threshold expressed as a count is 0.05 x 2,000 = 100 transactions.

Q40. Relate the position of the bend in an inertia curve to the value of k that should be reported.

A) The bend marks where extra clusters stop paying their way <-- correct
B) The bend marks the largest usable value of k
C) The bend marks the smallest possible inertia
D) The bend has no bearing on the choice of k
Answer: A. Before the bend each new cluster reduces inertia substantially; after it the reduction is slight, so the bend is the natural stopping point.

----- End of Sample Question Set -----